

Polymeric Insulation Materials for HVDC Cables: Development, Challenges and Future Perspective

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ABSTRACT

Extruded polymeric HVDC cables have drawn great attention in modern power systems. This paper reviews the history and development of the polymeric HVDC cables and summarizes the key technical problems in extruded HVDC cables. It is pointed out that two key issues should be solved. One is the electric field inversion within HVDC cable insulation which is caused by the temperature and electric field dependent DC volume resistivity of the polymeric insulation materials. The other is the space charge behavior under multi-fields coupling, including charge injection, transportation, accumulation and dissipation characteristics. The following aspects need to be particularly concerned in the future: the interaction between temperature, electric field, space charge and DC volume resistivity under multi-fields coupling; mechanisms of nanoparticles doping on enhancing the properties of polymeric insulation material; the long-term operation characteristics of nanodielectrics; collaborative optimal regulation methods and theories on the properties of polymeric insulation material; and recyclable insulation materials for future HVDC cables.

Index Terms — HVDC cables, polymeric insulation, electric field inversion, space charge, nanodielectrics, recyclable insulation material.

1 INTRODUCTION

POLYMERIC insulation materials have been very successfully used in the extruded high voltage alternating current (HVAC) cable underground power transmission for a long time. The first 110 kV extruded polymeric HVAC cable was developed in the 1950s and since then polymeric insulation material has been widely used in HVAC cables and gradually replaces traditional oil-filled (OF) cables and mass impregnated (MIND) cables. The first 500 kV extruded HVAC cable was used in Japan in 1988. But the development of high voltage direct current (HVDC) cables falls far behind the HVAC cables. The first HVDC cable was put into operation in 1954, connecting the Gotland Island and the Swedish mainland, which was ± 100 kV mass-impregnated cable [1]. Until 1999, the first commercial polymeric HVDC cable of ± 80 kV was put into service [2]. It has been reported that in Japan, ± 500 kV extruded HVDC cable has been developed, but is not used commercially [3].

Although the development history of extruded polymeric HVDC cable is not very long, it has shown great advantages in power transmission over HVAC cables and other kinds of HVDC cables such as OF and MIND cables [4], e.g.:

- (a) No reactive power transmission that ensures longer power transmission distance compared to HVAC cables;
- (b) Lower transmission loss compared to HVAC cables;

- (c) Conductor temperature can be raised compared to OF and MIND HVDC cables that makes the cable more compact under the same power rating;

- (d) Avoid the oil leaks which is a significant long-term potential environmental hazard in OF cables;

- (e) Joints for extruded HVDC cables are much simpler;

- (f) Lower maintenance cost compared to OF and MIND cables.

Not only the above mentioned advantages make polymeric HVDC cables develop rapidly, but the new demands of modern power transmission also contribute to the rapid development of polymeric HVDC cables. Safety, reliability, efficiency and environmental friendly have been the major restrictions in the development of modern power system [5]. So the demand of polymeric HVDC cables is increasing and its application range is being expanded. On one hand, offshore wind power and photovoltaic power transmission call for long distance and large capacity HVDC cable power transmission. On the other hand, underground HVDC cable power transmission can solve the contradiction between power transmission and cityscape during urbanization. Therefore, polymeric HVDC cables will be rapidly developed in the near future but the most important issue is the insulation material for polymeric HVDC cables.

In this paper, the development status of the polymeric insulation material for HVDC cables is reviewed and the required properties of the polymeric insulation are summarized. Also the problems in the development of polymeric HVDC cable insulation material,

especially space charge phenomena as well as the corresponding suppression methods are discussed. New attempts to develop recyclable polymeric insulation material are introduced. Finally, some important issues in the development of polymeric HVDC cable insulation materials are proposed. The aim of this paper is to summarize the state of the art of polymeric HVDC cable insulation materials and look into its future development.

2 HISTORY OF POLYMERIC INSULATION MATERIALS FOR CABLES

Polyethylene (PE) was firstly synthesized in 1933 and it can be classified into high density PE (HDPE), low density PE (LDPE) and linear low density PE (LLDPE) according to its molecular weight and degree of branching [6]. Because of the combination of excellent electrical properties, mechanical flexibility, reliability and low processing cost, LDPE has been used as the HVAC cable insulation material for about five decades. The first 225 kV LDPE insulated HVAC cable was installed in France in 1969 [7]. Since then, significant improvements have been achieved in the manufacturing of polymeric cables. The 400 kV LDPE insulated HVDC cables have been designed by Sagem Company in France, which have been put into operation in the HVDC power transmission project successfully.

Unfortunately, LDPE is not suitable for larger capacity power cables because its operation temperature is limited to 70 °C [8]. So, LDPE is crosslinked into crosslinking polyethylene (XLPE) to raise the operation temperature to about 90 °C [4] and the first use of XLPE in MV cables was in 1968 [1]. Now XLPE has been commonly used as HV cable insulation material. Radical reaction with organic peroxide (typically dicumyl peroxide, DCP) is one of the most widely used crosslinking methods to crosslink LDPE [4]. But the crosslinking process also introduces byproducts such as phenol, cumyl alcohol, acetophenone and α -methyl styrene, which would degrade the electrical properties and cause space charge accumulation. As a result, XLPE insulated cables often require degassing process to exclude these by-products during production [9-11]. Up to now, many manufacturers, such as ABB, Prysmian, Nexans, Viscas and LS cable have developed 500 kV XLPE insulated HVDC cables.

Ethylene-propylene rubber (EPR), a copolymer of ethylene and propylene has also been used as polymeric cable insulation material [12]. The properties of EPR can be controlled by the formulation of ethylene and propylene. When properly formulated, the properties of EPR can be superior to XLPE in some aspects. But EPR is generally expensive which limits its application in EHV cables [13]. Syndiotactic polypropylene (sPP), a kind of polypropylene synthesized by homogenous metallocene catalyst, has shown superior thermal and electrical properties to XLPE and is considered as the insulation material for next generation polymeric cables. Researchers have developed 22 kV sPP insulated cables and long-term transmission test was performed on the cables, which indicates that sPP insulated cable has a life span exceeding 30 years, but sPP is prohibitively expensive at present [14].

Recently, the concept of environmental protection and sustainable development is becoming increasingly important, so recyclable polymeric insulation material has attracted much attention [15]. As we know, crosslinking process makes XLPE a thermoset material which is difficult to be recycled and also causes much pollution and energy consumption during crosslinking and degassing process. Therefore, research on thermoplastic insulation materials are becoming a hotspot in the research and development of new polymeric insulation materials for HVDC cables. Polypropylene (PP) has shown great potential in the development because it is easy to be recycled and can be manufactured without any crosslinking process and byproducts and most importantly, it is cheap [15-19].

However, the poor mechanical properties limit the application of PP. Many researches have been focused on modifying the mechanical properties of PP and the most effective way is to blend with other thermoplastic elastomers, such as EPR, ethylene-propylene-diene monomer (EPDM), styrene-ethylene-butylene-styrene (SEBS) and polyolefin elastomer (POE) [20]. Hosier et al investigated the potential of propylene-ethylene copolymer and their blends as HVDC cable insulation material and the results indicate that these materials exhibit one or more desirable properties but none of them possesses all the advantages [21]. Zhou et al introduced POE to increase the flexibility of PP and MgO nanoparticles to suppress space charge accumulation and the obtained PP/POE/MgO nanocomposites possess both mechanical flexibility and excellent electrical properties [15, 18]. Although there is a distance for the practical application of PP in polymeric insulation HVDC cables, PP based insulation materials are promising. Recently, Prysmian Group announced that they have successfully developed the first prototype of 525 kV recyclable HVDC cable in 2016 [22]. They announced that this new kind of cable requires no chemical reactions to achieve the properties required for the long term electrical integrity and consequently no byproducts produced during the manufacturing process.

3 CHALLENGES IN HVDC CABLES

3.1 ELECTRIC FIELD INVERSION

3.1.1 ELECTRIC FIELD DISTRIBUTION IN HVDC CABLES

Electric field distribution in AC cables depends on the distribution of dielectric constant of the insulation material and the insulation geometry [23]. For polymeric insulation materials, the dielectric constant changes so slightly with temperature and electric field that can be ignored [24]. So the electric field distribution in AC cables can be easily calculated and the electric field distribution remains the same whether the cables are loaded.

But it is quite different in DC cables, because the electric field distribution in DC cables depends on the distribution of volume resistivity. Unfortunately, the DC volume resistivity of polymeric insulation materials changes by 2 to 3 orders of

magnitude in the temperature range of 20 °C to 90 °C. Figure 1 shows the DC volume resistivity of LDPE at various temperature and electric field [25].

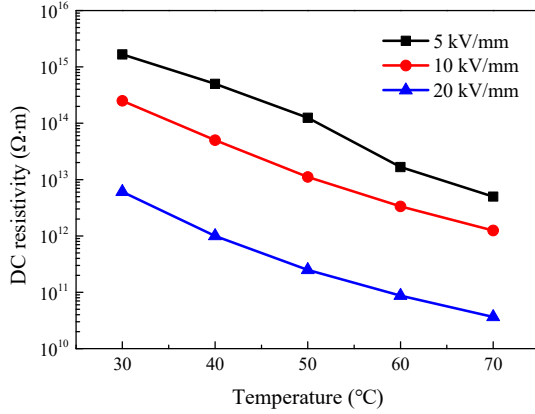


Figure 1. Temperature and electric field dependent DC volume resistivity of LDPE (data from [25]).

Temperature dependent resistivity means that the electric field distribution in DC cables also depends on the distribution of temperature. Namely, the electric field distribution in DC cables changes with the load current. The dependence of volume resistivity ρ on electric field E and temperature T can be described as [26]:

$$\rho = \rho_0 \exp(-\alpha T - \beta E) \quad (1)$$

where ρ_0 is the resistivity at reference temperature, α is the temperature coefficient, β is the electric field coefficient. So the electric field distribution $E(r)$ at radius r in DC cables can be calculated as [23, 27]:

$$E(r) = \frac{\delta U_0 (r/r_o)^{\delta-1}}{r_o [1 - (r_i/r_o)^\delta]} \quad (2)$$

$$\delta = \frac{\alpha W_c / 2\pi\lambda + \beta U_0 / (r_o - r_i)}{1 + \beta U_0 / (r_o - r_i)} \quad (3)$$

where λ is the thermal conductivity of the insulation material, r_o and r_i is the outer and inner insulation radius, U_0 is the voltage across the insulation and $W_c = I^2 R$ is the Joule loss of the conductor.

3.1.2 PROBLEMS WITH ELECTRIC FIELD INVERSION

It is clear that the electric field distribution in DC cables also depends on the load current. When the cable is under DC voltage but no load current, the electric field distribution in DC cables is also different from AC cables because of the electric field dependent volume resistivity. But it is certain that the highest electric field appears at the inner conductor in this condition. While for loaded DC cables, the highest electric field may appear at the outer insulation shield, which is so called electric field inversion. Figure 2 shows the electric field distribution in various insulation materials and the electric field inversion is obvious [24]. It should be noted that equation (1) is only an experimental result, and the theoretical dependence of DC volume resistivity on temperature and

electric field is not available until now. So it is very difficult to exactly calculate the electric field distribution in polymeric HVDC cables. Also, the electric field inversion which depends on the load current makes it difficult for the insulation design of HVDC cables.

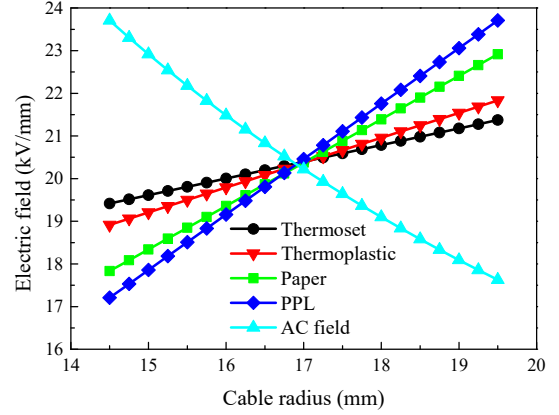


Figure 2. Electric field distribution in 100 kV DC cables with different insulations material under operation. The power loss W_c keeps the same (PPL for polypropylene/paper laminate) [24].

3.1.3 ATTEMPTS TO REDUCE ELECTRIC FIELD INVERSION

From equation (2), it can be concluded that higher thermal conductivity would contribute to reduce the temperature gradient through the insulation material and suppress electric field inversion. Also the DC volume resistivity of the insulation material should be insensitive to temperature and electric field.

3.1.3.1 MODIFICATION OF THERMAL CONDUCTIVITY

Polymeric insulation materials are generally thermally insulating with low thermal conductivity. So improving the thermal conductivity to reduce the temperature gradient is a good way to avoid electric field inversion. The most commonly used method of improving the thermal conductivity of polymeric insulation material is to introduce thermally conductive particles [28]. Various kinds of thermally conductive nanoparticles, such as Al_2O_3 , AlN , BN , Si_3N_4 and SiC have been introduced into polymeric insulation materials [29-31]. But when the thermally conductive nanoparticles introduced, other properties such as the mechanical properties and electrical properties may be affected, both positive and negative results have been reported in literature [28, 32]. So how to make a balance between these properties is an important issue in the future studies.

3.1.3.2 TEMPERATURE AND ELECTRIC FIELD COEFFICIENTS

Another way to reduce electric field inversion is to make the DC volume resistivity less sensitive to temperature and electric field. Some studies in nanodielectrics have found that introducing nanoparticles can reduce the volume resistivity at various temperature and electric field which means that the temperature and electric field coefficients of volume resistivity have been changed [25, 33-35]. Figure 3 is the

temperature and electric field dependent resistivity of LDPE and SiO₂/LDPE nanocomposites [25]. It is shown that the nanocomposites exhibit much lower resistivity and the temperature coefficient of resistivity in the nanocomposites decreases obviously at low electric field. The electric field coefficient also decreases at low temperature, but at high temperature, it is almost the same.

It should be noted that modification of the thermal conductivity and the temperature and electric field coefficient of electrical resistivity to achieve uniform electric field distribution in HVDC cable is a complicated work. If the thermal conductivity could let δ expressed in equation (3) equals to 1, then the electric field $E(r)$ becomes constant to r , which is expected in the design of HVDC cables.

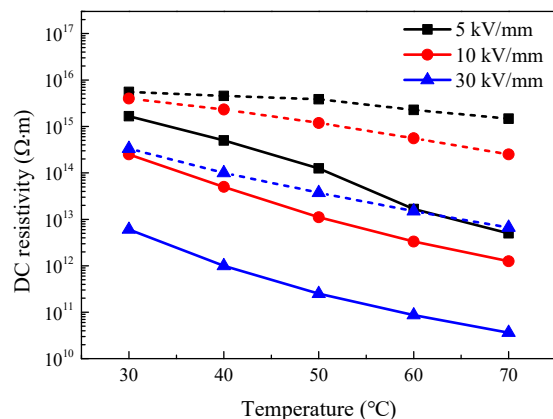


Figure 3. Temperature and electric field dependent DC resistivity of LDPE (data from [25]), solid lines for LDPE and dash lines for SiO₂/LDPE nanocomposites.

3.2 POLARITY REVERSAL AND SPACE CHARGE

3.2.1 POLARITY REVERSAL IN HVDC CABLES

The converter plays an important role in HVDC power system, there are typically two kinds of converters, current-sourced converter (CSC) and voltage-sourced converter (VSC). Earlier DC power system uses the CSC with thyristor valves in 1970's [36]. With the development of insulated-gate bipolar transistor (IGBT) valves, VSC HVDC was introduced in the late 1990's [37]. The major difference between CSC and VSC appears when there is a power flow reversal in the DC system. In CSC system, the power flow reversal is accomplished by changing the pole voltage polarity since the current flow direction cannot be reversed in CSC system. On the contrary in VSC system, the voltage polarity is always the same and the current flow direction is changed to reverse the power flow. Avoiding polarity reversal is extremely important for DC cable insulation, as it has been shown that the breakdown resistance can be reduced by 10 % on polarity reversal [4]. Although polarity reversal caused by power flow reversal does not exist in VSC system, it does not mean that there is no polarity reversal in VSC system because the electromagnetic transients can give rise to "temporary polarity reversal". For example, fast transient lightning overvoltage when the cables are terminated on an overhead line and switching transient at the DC substation.

As we all know, one of the most challenging issues in the HVDC cable insulation material development and insulation structure design is the space charge accumulation within the insulation material. The space charge accumulation could strongly affect the electric field distribution and cause local electric field distortion. If the space charge density is sufficient high, the induced local electric field strength would exceed the insulation breakdown strength and cause initial electric breakdown. Even if the local electric field is not so high, the space charge accumulation would gradually degrade the dielectrics and cause the formation of weak points. So the long-term reliability and life expectancy of the HVDC cables would be reduced. Figure 4 shows the space charge dynamics in LDPE under 330 kV/mm DC electric field at room temperature [38].

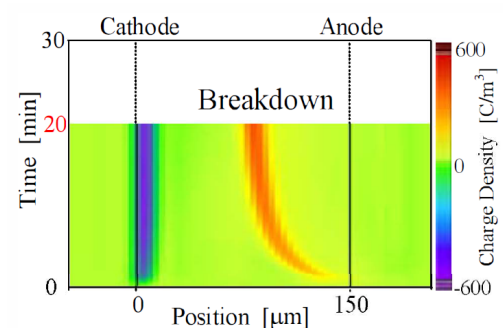


Figure 4. Time dependent space charge accumulation in LDPE under 330 kV/mm DC electric field at room temperature [38].

The worst condition about space charge is the formation of homocharges and then polarity reversal occurs [39, 40]. In this case, homocharges before polarity reversal become heterocharges after polarity reversal and the electric field would suddenly go higher and this has been confirmed by experimental data. Figure 5 shows the space charge and electric field distribution in film samples of LDPE before and after polarity reversal at 20 °C [41].

3.2.2 ORIGINS OF SPACE CHARGE

Considerable works on space charge phenomena in HVDC cable insulation materials have been done in the past, including the space charge formation, effect of temperature and temperature gradient, effect of different conductor-insulation interfaces [42-46] and the ways to suppress space charge accumulation [47-50]. The space charge accumulation within polymeric insulation material is a complicated process which is affected by the charge injection, transportation, extraction, trapping and detrapping process and depends on temperature, electric field and other factors [51]. In order to suppress space charge accumulation in HVDC cable insulation material, the origins of the space charge should be studied [52-55].

3.2.2.1 CHARGE INJECTION FROM THE ELECTRODES

Charge injection from the electrodes (i.e. electron injection from cathode and hole injection from anode) is often

considered as the origin of homocharge accumulation near the electrodes [56, 57]. The homocharge injection and accumulation would decrease the electric field near the injection electrode and increase the electric field near the opposite electrode. Usually, the charge injection depends on the potential barrier between electrodes and insulation material [57, 58]. Both electric field and temperature can increase the energy of the charge carriers and make it easier to overcome the barriers [59].

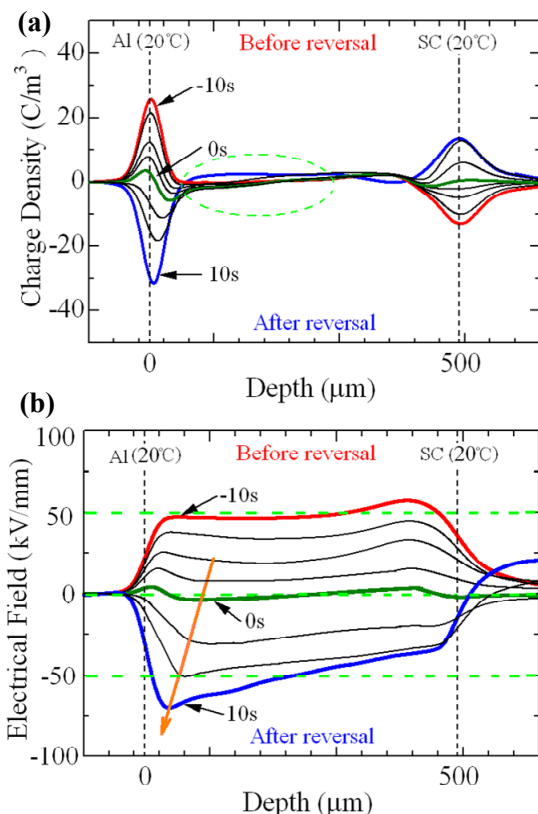


Figure 5. (a) Space charge and (b) electric field distribution in LDPE at room temperature during polarity reversal with the reversal time of 20 s [41].

3.2.2.2 FIELD-ASSISTED THERMAL IONIZATION OF IMPURITIES

Impurities are inevitably introduced by the raw materials and during the cable production process, and particularly the crosslinking byproducts would act as impurities in XLPE insulation. These impurities would thermally ionize into positive ions and negative electrons under the assistance of DC electric field [60]. Most of the ions and electrons would recombine in a short time. The remaining electrons and ions would migrate to the anode and cathode, respectively and may be trapped by the charge traps near the electrode which forms heterocharges [61]. Space charge from electric-assisted thermal ionization would increase with the applied electric field.

3.2.2.3 PERMITTIVITY AND CONDUCTIVITY INHOMOGENEOUS

As discussed above, the permittivity and conductivity of the polymeric insulation material is the function of radius r

because of the temperature gradient in loaded HVDC cables. According to the electromagnetic theory, the space charge density ρ in steady DC current field with the current density j can be given as:

$$\rho = j \cdot \nabla(\epsilon_r \epsilon_0 / \sigma) \quad (4)$$

where ϵ_0 is the vacuum permittivity, ϵ_r is the relative permittivity of the insulation material and σ is the conductivity of the insulation material.

For polymeric insulation materials, the conductivity remarkably depends on temperature and electric field, while the corresponding variation in permittivity is much smaller. So the charge density expressed in equation (4) does not equal to 0. References [62-64] found that space charge originated from the inhomogeneous of permittivity and conductivity in 250 kV XLPE insulated cable is in the order of 10 nC/cm³.

3.2.3 ATTEMPTS TO SUPPRESS SPACE CHARGE

Since space charge accumulation in polymeric HVDC cable insulation material is an essential problem in the development of high performance insulation materials. Space charge suppression is a hot topic in both scientific research and industrial application [65-67]. Generally, there are three ways to suppress space charge accumulation in polymeric insulation materials, (a) introducing inorganic nanoparticles, (b) grafting with special functional groups, (c) blending with other polymers.

3.2.3.1 INTRODUCING INORGANIC NANOPARTICLES

Introducing inorganic nanoparticles into polymer insulation materials is especially called nanodielectrics, which is very popular in scientific research and is regarded as the new generation polymeric insulation material. Since the concept of nanodielectrics was proposed by T. J. Lewis in 1994 [68], it has shown great favorable properties [69-71]. Various kinds of inorganic nanoparticles have been studied, such as silica, magnesia, titanium oxide, alumina, zinc oxide, layered silicates (clays), montmorillonite, zeolite, boron nitride and so on [72-75]. Table 1 summarizes the effects of different nanoparticles on the properties of polyethylene [76-80].

Due to the high specific surface area of nanoparticles, a few percent addition of nanoparticles can significantly affect the dielectric properties, especially space charge suppression in polymeric material. Figure 6 is the space charge distribution in LDPE and MgO/LDPE nanocomposites under 100 kV/mm DC electric field, which shows that the introduction of MgO nanoparticles can effectively suppress space charge accumulation [81]. The superior properties of nanodielectrics are attributed to the numerous interfaces between nanoparticles and polymer matrix and there are also some models to explain the properties of nanodielectrics, such as the electric double layer model proposed by T. J. Lewis [82] and the multi-core model proposed by T. Tanaka [83].

3.2.3.2 GRAFTING WITH SPECIAL FUNCTIONAL GROUPS

Recently, there are attempts to modify the properties of polymeric materials by grafting with special functional chemical

Table 1. Summary of the exist results in LDPE nanocomposites.

Material	Filler content	DC breakdown strength	Dielectric constant	Dielectric loss	Resistivity	Space charge
LDPE/SiO ₂ [76]	5 wt%	decreased	slightly decreased	slightly decreased	increased 13.3 times	heterocharge suppressed
LDPE/MgO [77]	0.2, 0.5, 1, 2, 5, 10 phr	38 % increased	---	---	increased 15 times	space charge suppressed at high electric field
LDPE/ZnO [78]	0.1, 0.5, 1, 3, 5, 7, 10 wt%	---	---	---	increased 4 times	space charge suppressed at 7 wt%
LDPE/TiO ₂ [79]	2, 10 wt%	---	10 % increased	---	increased 2 times	space charge suppressed at 10 wt%
LDPE/Al ₂ O ₃ [80]	0.1, 0.2, 0.5, 1 wt%	obviously decreased	16 % decreased	---	increased 3.4 times	hetero space charge suppressed

groups. The special functional groups are usually polar groups, which would introduce charge carrier traps to suppress space charge accumulation [84-86]. It was found that space charge accumulation can be suppressed to 50-70 % in XLPE when grafted with maleic anhydride (MAH) [87]. Further researches indicated that introduction of polar groups, e.g. carbonyl, hydroxyl, nitro, cyano and aromatic ring can effectively suppress space charge accumulation and increase the volume resistivity of PE and other α -olefin copolymers [88]. Our recent work has shown that grafting MAH with PP can suppress space charge accumulation and increase DC breakdown strength and volume resistivity [89]. Patent of Prysmian pointed out that grafting with unsaturated fatty acids in PE can suppress space charge accumulation and enhance the electrical properties of HVDC cables under polarity reversal [90]. ABB revealed that grafting with 1 wt% dimethylaminopropyl methacrylamide (DMPMA) can suppress space charge accumulation in XLPE [91].

HVDC cable insulation material depends complicatedly on temperature, electric field and the accumulation of space charge. Considering the above conclusions and previous studies, the optimal properties of polymeric insulation material for HVDC cables can be suggested as follows:

- Stabilized volume resistivity to the variation of temperature, electric field, polarity reversal and polarization time.
- Large thermal conductivity to reduce the temperature gradient within the insulation of loaded HVDC cables.
- Suppressed space charge accumulation and preferably no space charge accumulation or large space charge dissipation rate.
- High DC breakdown strength and insensitive to temperature and polarity reversal.
- High breakdown strength under DC voltage superposed with impulse voltage.

Realization of these characteristics is always a hard work.

4 FUTURE PERSPECTIVE

4.1 SPACE CHARGE BEHAVIOR UNDER MULTI-FIELDS COUPLING

Space charge greatly restricts the development and safe operation of polymeric HVDC cables. So investigations on space charge related phenomena have been regarded as the key foundation for the design and evaluation of polymeric HVDC cable insulation material. Therefore, space charge generation, transportation, accumulation and dissipation under multi-fields coupling condition is a key issue to be solved.

In polymeric HVDC cable insulation materials, the space charge behavior is influenced by the electrical conductivity, thermal conductivity, charge traps distribution and electric field strength. Also the electric field distribution is influenced by the space charge accumulation. So the nonlinear relationship between these parameters and the external excitation (temperature and electric field) makes the causality of the material properties, electric field distribution, temperature distribution and space charge distribution coupled with each other. Then how to achieve the collaboration and accordance of the physical properties of insulation material and external excitation is very important for the design of polymeric HVDC cables.

It has shown that space charge injection and extraction are affected by the electrodes temperature, higher temperature

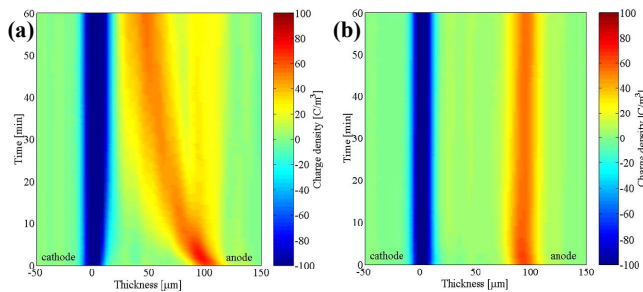


Figure 6. Space charge distribution in (a) LDPE and (b) MgO/LDPE nanocomposites under 100 kV/mm DC electric field at room temperature [81].

3.2.3.3 BLENDING WITH OTHER POLYMERS

K. S. Suh et al investigated the PE blended with methacrylic ionic compound and the results indicate that the space charge can be suppressed and distributed more uniform [92]. Researches by T. Suzuki used ionomer to blend with XLPE and HDPE, respectively and the space charge was measured by thermally stimulated current (TSC) measurements. It is shown that the space charge accumulation is reduced and its distribution pattern is changed [93, 94]. It is also shown that blending different thermoplastic polymers can also suppress space charge accumulation by introducing interfaces between different polymers [95, 96].

3.3 REQUIRED PROPERTIES FOR POLYMERIC HVDC CABLE INSULATION MATERIAL

From the analysis above, the electrical properties of polymeric

could assist the charge injection and extraction. So space charge accumulation under high temperature should be further studied. Figure 7 is an example of space charge accumulation in XLPE and SiO₂/XLPE nanocomposites [97]. It is shown that space charge injection under high temperature is more obvious and there are heterocharges accumulation near the anode. But for the nanocomposites, heterocharges accumulation under both low and high temperature is suppressed.

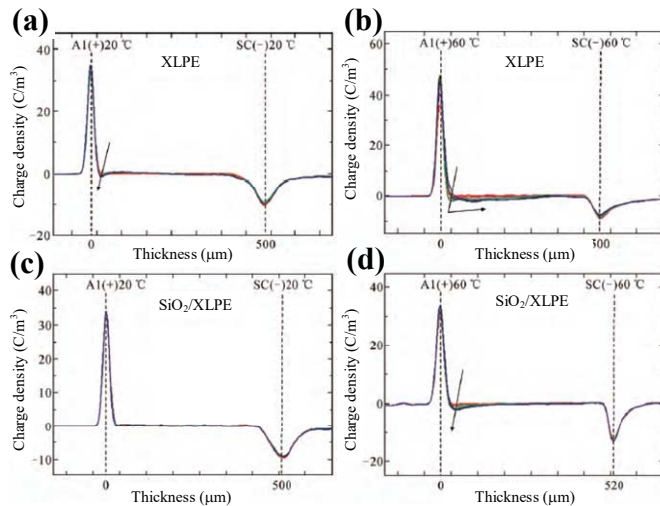


Figure 7. Space charge distribution in XLPE at (a) 20 °C, (b) 60 °C and SiO₂/XLPE nanocomposites at (c) 20 °C, (d) 60 °C [97].

However, space charge behavior under temperature gradient is different from that of under steady temperatures, because the injection and extraction characteristic differs with the temperature of the electrodes. Studies of space charge under temperature gradient is closer to the actual operation condition of the HVDC cables. The charge injection and extraction at the high temperature electrode would be easier and for the low temperature electrode, it is harder. So there are usually heterocharges accumulation at the low temperature electrode in the polymeric insulation material under temperature gradient. Figure 8 is an example of space charge and electric field distribution in LDPE film samples under steady temperature of 20 °C and temperature gradient of $\Delta T = 40$ °C [41].

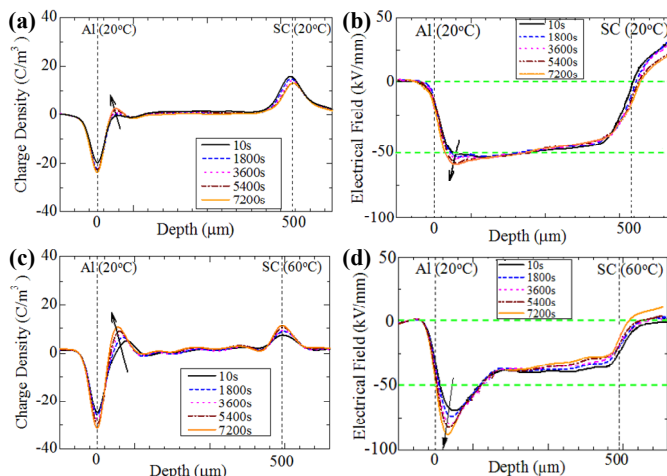


Figure 8. Space charge and electric field distribution in LDPE at 20 °C (a, b) and 20-60 °C temperature gradient (c, d) [41].

First of all, the understanding of space charge is based on the measurement technology. The most widely used methods of space charge measurement are laser-intensity-modulation-method (LIMM), the thermal step method (TSM), the pressure-wave-propagation (PWP) or laser-induce-pressure-pulse (LIPP) method, and the pulsed-electro-acoustic (PEA) method [98, 99], details about these methods can be found in [100]. Although these methods are widely used, the spatial and temporal resolution should be increased to quantitatively describe the dynamics of space charge development. Also, space charge behavior under multi-fields coupling, e.g. electric field, temperature gradient, mechanical stress, should be comprehensively studied. A deep understanding of the effect of temperature and temperature gradient, electric field, mechanical stress and other factors on the formation and development of space charge and the interaction between space charge and electrical conductivity should be established.

4.2 THEORY AND METHOD OF MULTI-PROPERTIES COORDINATED MODIFICATION

Nanodielectrics have shown superior properties in many aspects such as electrical breakdown strength, space charge suppression, PD resistance, electrical treeing suppression, thermal conductivity, electrical resistivity and temperature tolerance, which have shown great potential in polymeric HVDC cable insulation material application [101]. But most of the studies focused on only some of the properties and how to achieve the optimal performance of all the properties is still a great challenge. Therefore, the effect of different nanoparticles on different properties of the polymeric insulation materials should be comprehensively studied to establish the methods of tailoring different properties [102]. Based on the methods developed, the optimal modification could be achieved by designing the formulation and nanostructure inside the material.

The superior properties of nanodielectrics have been regarded as the properties of the interfaces between nanoparticles and polymer matrix. The interface is considered to have unique physical and chemical properties due to its special formation mechanism. Although there have been several models to describe the interfaces, they cannot explain all the experimental results [82, 83, 85]. Difficulties in the interfaces research come from the very small spatial scales of the interfaces, which is far less than the resolution of most conventional test methods. Development of accurate detection method at nanoscale is an essential step in the research of interfaces.

Recently, electrostatic force microscopy (EFM) has shown the potential to be used to investigate the dielectric properties of interfaces at nanoscale [103]. T. S. Jespersen et al used EFM to characterize the surface structure and dielectric properties of PMMA/carbon nanotube composites [104]. M. Labardi et al conducted the local dielectric property measurements of PVAc/MMT composites via EFM, and found that the dielectric peak of the interface broadened, which proves that the mobility of PVAc chain segments in the interface would be influenced by MMT sheets [105]. This method may be helpful to uncover a corner of the interface performance between nanoparticles and polymer.

4.3 RECYCLABLE INSULATION MATERIALS AND NEW DOPANTS

Environmental protection and sustainable development have been even more important than ever, so recyclable polymeric insulation materials are becoming popular. Crosslinking byproducts in traditional XLPE insulation would cause space charge accumulation and reduce the life of the cables. Although introduction of nanoparticles can suppress space charge accumulation, the problems with crosslinking byproducts cannot be completely solved. Thermoplastic materials, which need not to be crosslinked, provide a new solution. Polypropylene based materials have shown great potential in HVDC cable insulation application and are being extensively studied [15-19]. Adopting thermoplastic insulation materials in HVDC cables could greatly simplify the production procedure, save energy consumption and do not need to consider the impact of crosslinking byproducts. So recyclable insulation materials would be the future alternative to XLPE.

Traditional nano-fillers are usually spherical or spheroidal inorganic oxide nanoparticles, but recent advances in nanodielectrics have shown that some nanoparticles with special shapes or properties can exhibit superior properties. M. Jarvid et al found that C₆₀ and its derivatives [6, 6]-phenyl-C₆₁-butyric acid methyl ester (PCBM) could act as the voltage stabilizer to increase the breakdown strength and inhibit the formation of electrical trees in XLPE because of the high electron affinity and low ionization energy of C₆₀ and PCBM, which scavenge the high-energy electrons [106]. Recent work by Y. Yang et al designed an interesting nano-filler, mesoporous silica MCM-41 loaded with polyethylenimine (PEI), and the loaded PEI would absorb the degradation products of polymer and increase the compatibility of PP and MCM-41 [107]. The MCM-41-PEI would act as the degradation inhibitor to enhance the durability and stability of polymer dielectrics, especially at high temperature. Some nanosheets also exhibit superior effect on modifying the properties of polymer. Q. Li et al found that boron nitride nanosheets (BNNSs) could improve the breakdown strength and volume resistivity of polymer dielectrics, particularly at high temperature, which is desirable in industrial application. Also the thermal conductivity can be dramatically increased with the incorporation of BNNSs [108]. So developing new kinds of nano-dopants and designing the structure or physical and chemical properties of the nanoparticles would be a promising way to prepare high performance nanodielectrics.

5 CONCLUSIONS

This paper reviews the history, development, challenges and perspective of polymeric HVDC cable insulation materials. Future power transmission will be toward to high capacity, long distance, high economic efficiency, high reliability and low environmental impact. Therefore HVDC power transmission, especially HVDC cable power transmission is highly desirable. So further researches on polymeric HVDC cable insulation materials are needed. For polymeric HVDC cable insulation materials, there are basically two essential problems. One is the

electric field inversion caused by the temperature and electric field dependent DC volume resistivity. The other is the space charge accumulation which affects the electric field distribution and causes local electric field distortion. The difficulties in material performance optimization come from the complicated interaction between temperature, electric field, space charge accumulation, electrical resistivity, mechanical stress and so on. So studies on the relationship between these parameters are extremely important. Nanodielectrics have shown great potential but the theory and method of multi-properties coordinated modification should be further studied. Researches should also be focused on the development of recyclable polymeric HVDC cable insulation materials.

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